Introduction to Security

Sam Ogden

DRAFT

This slide deck is still under active development.
Expect changes before it is finalized.

Lecture Objectives

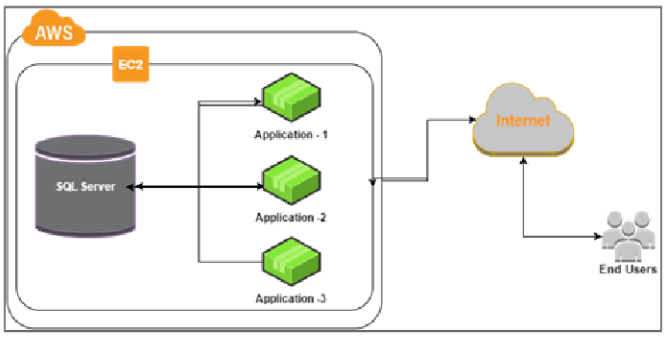
- After this lecture, you should be able to:
- Explain why security is an essential consideration for operating systems
- Discuss the goals of security
- Core security concepts

What we've done so far

- So far we've virtualized resources:
- Limited Direct Execution for processes
- Paging for memory
- That keeps processes apart from each other
- But isolation alone does not answer how shared resources should be controlled
- Is that really enough?

Resource Sharing is still important

- Not all applications can share memory with just threads
- Sometimes we need to communicate between processes
- Should all processes have equal footing?



What might we want to protect?

- Security starts by naming the assets that matter
- Process memory
- Passwords, personal information, execution pathways
- Files
- User passwords are stored in a file. Who should be able to write?
- I/O devices
- e.g. Who should be able to print?
- e.g. Who should be able to send network messages?
- Resource decisions!

We need the OS to help us with this!

The OS already controls the resources we need to protect

That makes it the natural place to enforce security policy

OS Responsibilities

- The OS owns memory management and sharing, so it must protect each process's memory
- The OS starts, pauses, and terminates processes, so it must make reasonable choices
- Resources are made available by the OS, so it must allocate them safely

Security Goals

- **Confidentiality**: keep information secret
- **Integrity**: keep information correct
- **Availability**: keep the system up and running

Example: A cryptominer virus

- Mines crypto using your GPU, without your permission
- Increases your power bill
- Reduces your game's FPS

What goals does this circumvent?
Availability

Security design is about tradeoffs

- The goal is not just to block attacks
- The goal is to protect the system without making it unusable
- Good security design reduces damage when something goes wrong

Design Principles

Keep it simple

- Economy of mechanism
- Open design

Enforce policy carefully

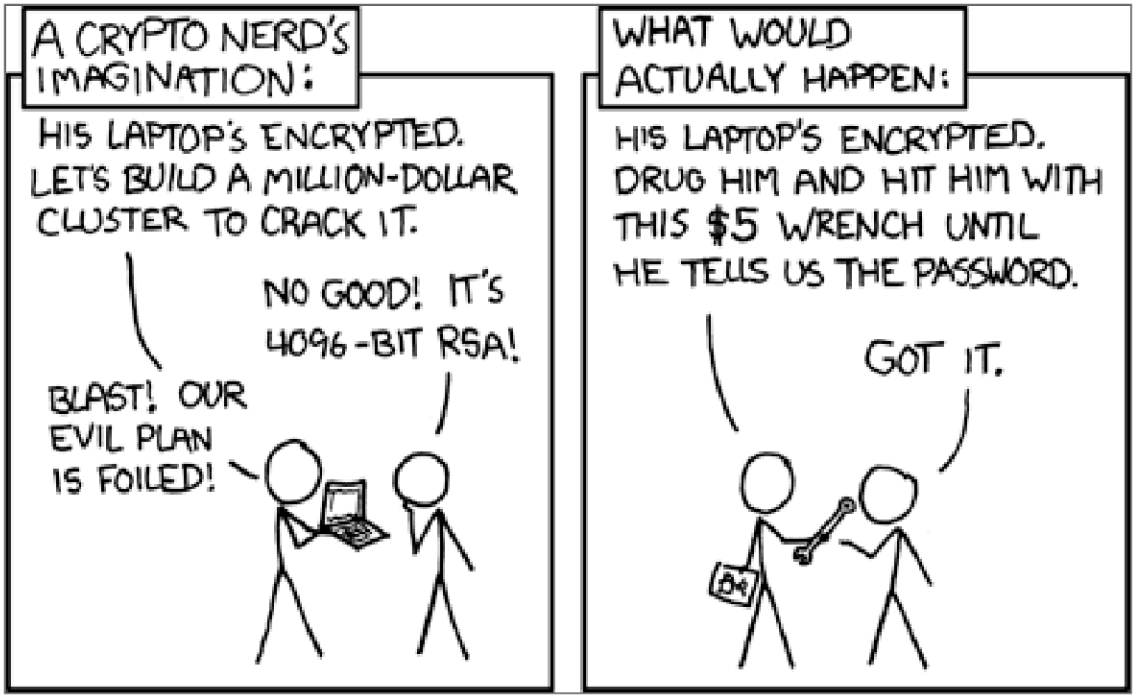
- Fail-safe defaults
- Complete mediation
- Separation of privilege

Limit blast radius

- Least privilege
- Least common mechanism
- Acceptability

Encouraging good security

- Weakest links will break first
- Breaking a cryptographic hash is really hard
- Breaking “password” (i.e. the literal word as a password) is really easy



System Calls - Not all are equal

- `exit()`
- `getpid()`
- `getppid()`
- `time()`
- `wait()`
- `fork()`
- `mmap()`
- `mprotect()`
- `socket()`
- `recv()`

Summary

- Security is essential to any system
- Virtualization and sharing are useful, but they create policy decisions that the OS must enforce
- The security goals are confidentiality, integrity, and availability
- Good security comes from simple mechanisms, careful checks, and designs people can actually use
- In practice: focus on what works, and make sure it still works when things go wrong

Speaker notes